Physics 210 Tutorial # 1 Multiple Choice Problems

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. The position of a particle moving along the *x* axis is given by $x = 6.0t^2 - 1.0t^3$, where *x* is in meters and *t* in seconds. What is the position of the particle when it achieves its maximum speed in the positive *x* direction?
a. 24 m
b. 12 m
c. 32 m
d. 16 m
e. 2.0 m

____ 2. The position of a particle as it moves along the *x* axis is given by $x = 15e^{-2t}$ m, where *t* is in s. What is the acceleration of the particle at $t = 1$ s?
a. 22 m/s
b. 60 m/s
c. 8.1 m/s
d. 15 m/s
e. 35 m/s

____ 3. Cart A, of mass *m*, starts from rest and travels in a straight line with acceleration *a*. It traverses a distance *x* in time *t*. Cart B, of mass 4*m*, starts from rest and travels in a straight line with acceleration $\frac{a}{2}$. At time *t* it has traversed the distance
a. $\frac{x}{4}$.
b. $\frac{x}{2}$.
c. *x*.
d. 2*x*.
e. 4*x*.

Instructions: On occasion, the notation $\vec{\mathbf{A}} = [A, \theta]$ will be a shorthand notation for $\vec{\mathbf{A}} = A\cos\theta\hat{\mathbf{i}} + A\sin\theta\hat{\mathbf{j}}$.

____ 4. If $\vec{\mathbf{A}} = 12\hat{\mathbf{i}} - 16\hat{\mathbf{j}}$ and $\vec{\mathbf{B}} = -24\hat{\mathbf{i}} + 10\hat{\mathbf{j}}$, what is the magnitude of the vector $\vec{\mathbf{C}} = 2\vec{\mathbf{A}} - \vec{\mathbf{B}}$?
a. 42
b. 22
c. 64
d. 90
e. 13

____ 5. A vector, $\vec{\mathbf{B}}$, when added to the vector $\vec{\mathbf{C}} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ yields a resultant vector which is in the positive *y* direction and has a magnitude equal to that of $\vec{\mathbf{C}}$. What is the magnitude of $\vec{\mathbf{B}}$?
a. 3.2
b. 6.3
c. 9.5
d. 18
e. 5

____ 6. At $t = 0$, a particle leaves the origin with a velocity of 9.0 m/s in the positive y direction and moves in the xy plane with a constant acceleration of $(2.0\mathbf{i} - 4.0\mathbf{j})\text{m/s}^2$. At the instant the x coordinate of the particle is 15 m, what is the speed of the particle?

a. 10 m/s
b. 16 m/s
c. 12 m/s
d. 14 m/s
e. 26 m/s

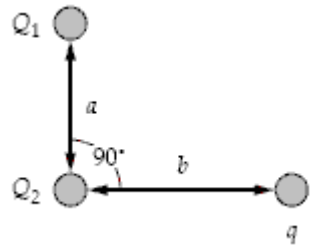
____ 7. A rock is projected from the edge of the top of a building with an initial velocity of 12.2 m/s at an angle of 53° above the horizontal. The rock strikes the ground a horizontal distance of 25 m from the base of the building. Assume that the ground is level and that the side of the building is vertical. How tall is the building?

a. 25.3 m
b. 29.6 m
c. 27.4 m
d. 23.6 m
e. 18.9 m

____ 8. Each of two small non-conducting spheres is charged positively, the combined charge being 40 μC. When the two spheres are 50 cm apart, each sphere is repelled from the other by a force of magnitude 2.0 N. Determine the magnitude of the smaller of the two charges.

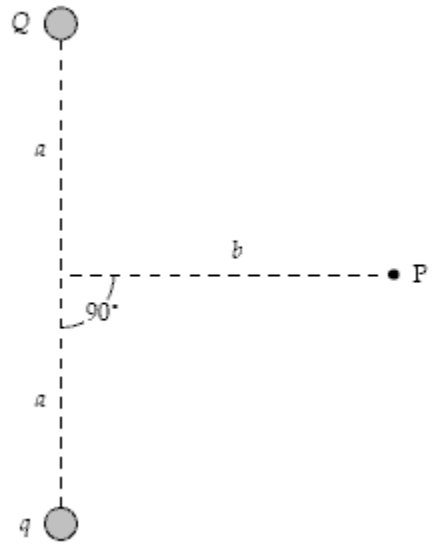
a. 1.4 μC
b. 1.1 μC
c. 2.0 μC
d. 3.3 μC
e. 17 μC

____ 9. If a = 3.0 mm, b = 4.0 mm, Q_1 = 60 nC, Q_2 = 80 nC, and q = 24 nC in the figure, what is the magnitude of the total electric force on q?



a. 2.7 N
b. 1.9 N
c. 2.3 N
d. 1.5 N
e. 0.52 N

____ 10. If a = 60 cm, b = 80 cm, Q = –6.0 nC, and q = 3.0 nC in the figure, what is the magnitude of the electric field at point P?



a. 71 N/C
b. 56 N/C
c. 60 N/C
d. 53 N/C
e. 67 N/C

Physics 210 Tutorial #1 Practise Problems
Answer Section

MULTIPLE CHOICE

1. ANS: D PTS: 1
2. ANS: C PTS: 1
3. ANS: B PTS: 1
4. ANS: C PTS: 1
5. ANS: A PTS: 1
6. ANS: A PTS: 1
7. ANS: D PTS: 1
8. ANS: A PTS: 1
9. ANS: D PTS: 1
10. ANS: D PTS: 1